



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

DISTRIBUTED CLOUD SECURITY [IT-4002]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) You are part of a cybersecurity team tasked with improving user authentication mechanisms for a cloud-based enterprise application that handles sensitive customer data. Based on your understanding of modern authentication techniques, apply appropriate strategies to enhance user authentication and ensure a high level of data security. (5)
 - A)
 - B) Apply the steps of the Needham-Schroeder (NS) protocol to demonstrate how secure communication is established between two parties and represent the process with a clear and accurate diagram. (5)
- 2) Apply appropriate access control mechanisms to regulate and restrict access to sensitive data and resources in specific security scenarios and justify your choices. (5)
 - A)
 - B) Solve common cybersecurity issues in modern enterprises by applying Identity and Access Management (IAM) frameworks that ensure secure and appropriate resource access. (5)
- 3) Apply your understanding of network security to explain how the Ping of Death attack can affect system stability. Demonstrate how to recognize its symptoms in a network environment and implement appropriate security measures to prevent and mitigate its impact. (5)
 - A)
 - B) Apply Shared Responsibility Model in cloud computing define the security roles and responsibilities of cloud service providers (CSPs) and customers? (5)
- 4) Apply your understanding of common cloud security risks by identifying the top seven threats in a cloud environment, and demonstrate how appropriate mitigation strategies can be used to address each risk effectively (5)
 - A)
 - B) An IT team identifies gaps in data access control and audit logging. Use the CIA triad to evaluate the risks associated with these gaps and propose solutions that also align with regulatory compliance. (5)
- 5) A company is planning to implement server virtualization to improve resource utilization and system isolation. As a virtualization consultant, you are tasked with recommending the appropriate hypervisor type. Analyse the key differences between Type 1 and Type 2 hypervisors in terms of architecture, performance, and security, and discuss the advantages of each. (5)
 - A)

- B) Your organization is considering deploying VMware ESXi to virtualize its server infrastructure for better efficiency and security. As a virtualization specialist, you are asked to analyse its suitability. Analyse the key features, architecture, and security enhancements of VMware ESXi, and describe how these aspects support robust, scalable, and secure virtualization in an enterprise environment. (5)

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